

Employment

Graduate Research Assistant	NC State University	Apr 2025 – Present
------------------------------------	----------------------------	---------------------------

- Collaborated with plant biologists and geneticists to deploy cloud-native genomic data platform on GCP (Cloud Run) with JBrowse 2 genome browser, enabling 100+ researchers to visualize 500GB+ genome datasets.
- Improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.

Product Engineer Intern	Isomer AI	May 2025 – May 2026
--------------------------------	------------------	----------------------------

- Shipped per-tenant AI workflow platform for insurance carriers: Temporal.io durable orchestration, LLM risk signal detection across Claude/GPT-4o/Gemini, Auth0 RBAC, MCP servers; 30+ deployments.
- Built MCP servers (Python + TypeScript); Playwright + Tailscale VPN for customer-internal browser automation.

Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
----------------------------------	-----------------------------	----------------------------

- Partnered with healthcare data teams to build 5+ AI workflows including health claim automation using OCR, NLP, and ML pipelines; improved document processing accuracy by 25% across 10K+ daily medical claims.
- Automated AWS and GCP CI/CD pipelines; integrated Grafana monitoring and led model validation and red-teaming sessions.

AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
---------------------	------------------	----------------------------

- Led cross-functional SDLC for data-driven enterprise ticket analysis using NLP classification models; worked with support teams to reduce resolution time by 15% and eliminate 2,000+ monthly manual triages.
- Built Kafka + Spark + Hadoop pipelines with data engineers; cut ticket-processing latency by 20% at peak load.

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
--------------------	--	----------------------------

- Master of Science in Computer Science; GPA: 3.5/4.0. Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering, Programming Languages, Computer Networks, Distributed Systems.

Mumbai, India	Mumbai University	Aug 2020 – May 2024
----------------------	--------------------------	----------------------------

- Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infrastructure, DS & Algorithms, Machine Learning, Blockchain, Computer Networks, Database Systems.

Projects

CryptoKnight	Trustless AI Trading Agent	2025
---------------------	-----------------------------------	-------------

- Built autonomous crypto trading agent using ERC-4337 Account Abstraction: scoped session keys enforce on-chain trade policies (caps, allowances, expiry); Python agent farms Polymarket; zero private key exposure.
- Alchemy Account Kit, viem, wagmi on Ethereum Sepolia; self-custody smart account vault; 0 private keys shared with agent.

KubeServe	Kubernetes LLM Inference Platform	2025
------------------	--	-------------

- Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N replicas), Kopf CRD operator for multi-provider LLM serving; circuit breaker + EWMA load balancing.
- Deployed via Helm + ArgoCD GitOps on Kubernetes; OTel + Prometheus + Grafana observability stack; 88 pytest tests.

AlgoPulse	Full-Stack Agentic AI Platform	2025
------------------	---------------------------------------	-------------

- Built agentic AI orchestration platform on Temporal.io and Kafka: agent loop with multi-step reasoning, tool invocation, and retry logic; sustained 2,000 async inference requests per second.
- Deployed on AWS with auto-scaling groups; k6 load tested to 10,000 concurrent users; sub-100ms leaderboard latency.

Languages and Technologies

- **Languages:** Python, TypeScript, JavaScript, SQL, Bash
- **Data & Streaming:** Kafka, PostgreSQL, MySQL, Redis, DuckDB, Parquet, Testcontainers
- **Infra & DevOps:** Kubernetes, Docker, AWS, GCP, Linux, Prometheus, Grafana, GitHub Actions, ArgoCD, Helm, OpenTelemetry
- **Crypto & Web3:** ERC-4337, Session Keys, Ethereum, viem, wagmi, Alchemy Account Kit, Polymarket, Sepolia

Additional Experience and Awards

- **IEEE Published:** Co-authored and presented research on Computer Vision-Based Cybersecurity Threat Detection at the Springer icSoftComp 2023 international conference, reaching an audience of 500+ global attendees.
- **Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; built 3 full-stack MVPs end-to-end in 48-hour competitive sprints.
- **GDSC Lead:** Led 4+ tech events as Google Developer Student Club Lead, reaching 500+ students across hackathons and conferences.
- **Technical Writing:** Published articles on LLM model collapse, AI agent security, and audio AI at kulkarniaditya.com/blog.